

Challenges

Relational Degradation: Missing views distort cross-view relational structures, not just feature entries.

Imputation Artifacts: Generative methods incur high computational costs and introduce synthetic noise.

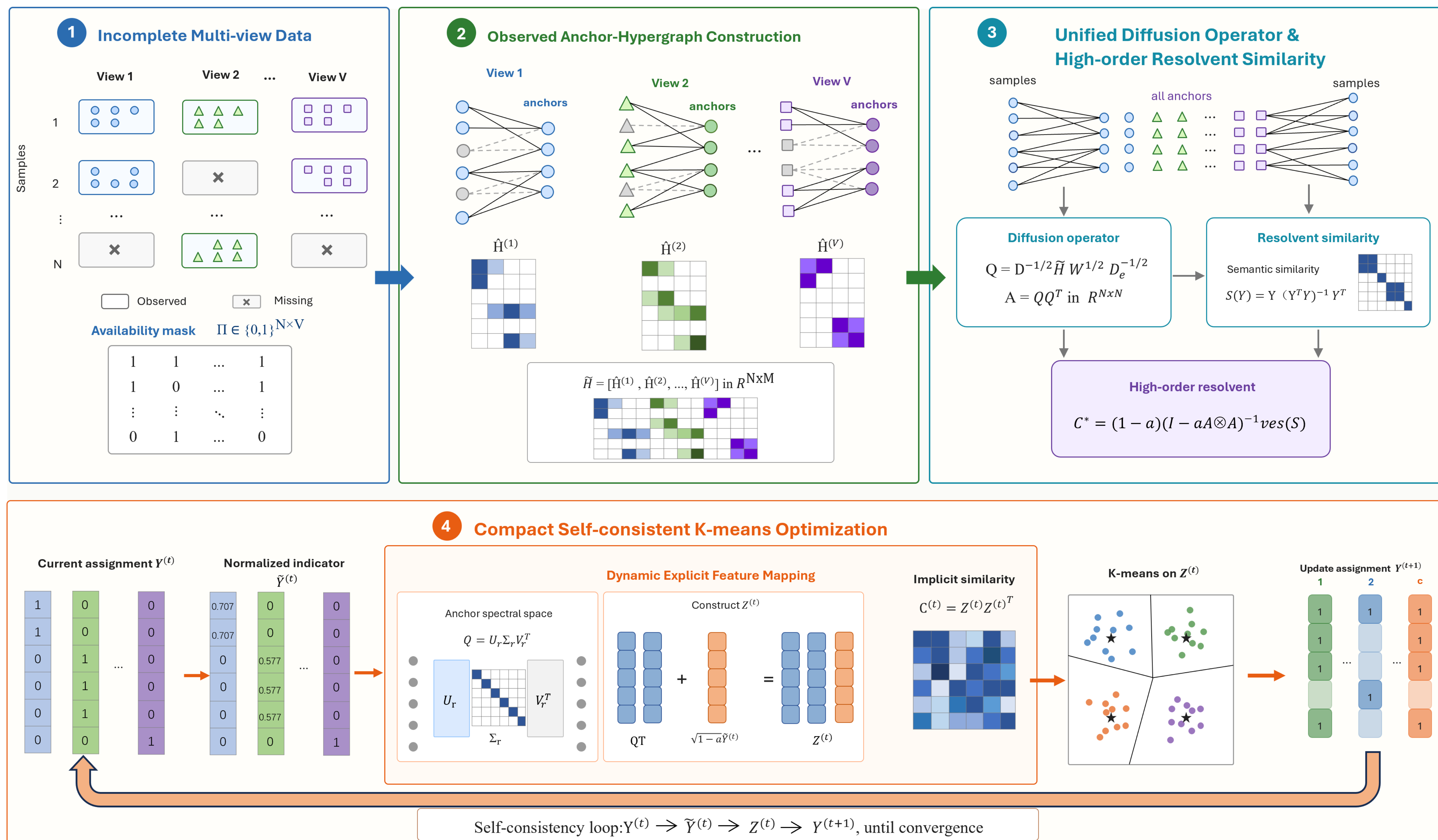
Shallow Fusion & Scalability:
Standard first-order graphs struggle with high missing rates and suffer from $O(N^2)$ bottlenecks.

Motivation

From a Graph Signal Processing (GSP) perspective, standard first-order fusions are shallow low-pass filters with limited spectral selectivity.

Rather than explicitly reconstructing data, we reformulate Incomplete Multi-View Clustering (IMVC) as a high-order spectral filtering problem to refine degraded structures using only partial observations.

Methods



Theoretical Analysis

Tunable Spectral Selectivity: The resolvent filter mathematically amplifies latent consensus modes ($SSR > 1$) while suppressing view-specific residual noise.

Positive Path-Supported Diffusion: Multi-hop anchor paths naturally construct the similarity matrix, entirely bypassing the synthetic noise introduced by artificial feature imputation.

Local Subspace Refinement:
Conditional local subspace contraction guarantees theoretical stability and strictly bounded error during iterative optimization.

Conclusions

Paradigm Shift: Successfully reformulates Incomplete Multi-View Clustering (IMVC) from a data imputation paradigm to a principled spectral filtering approach.

Robustness & Efficiency: Achieves robust structural refinement under severe missingness, maintaining exact linear $O(N)$ scaling for a fixed anchor budget.

Key Insight

Imputation-Free Hypergraph: We construct a unified anchor-induced hypergraph strictly from observed sample-anchor relations.

Resolvent Spectral Filter: We introduce a high-order resolvent diffusion operator that amplifies true consensus modes and suppresses view-specific variations.

Linear Scalability: Our implicit solver avoids dense matrix materialization, enabling exact Euclidean K-means with $O(N)$ complexity for a fixed anchor budget.

Results

Dataset	Method	ACC					NMI					ARI				
		0.1	0.3	0.5	0.7	0.9	0.1	0.3	0.5	0.7	0.9	0.1	0.3	0.5	0.7	0.9
UCiDDigits	PIC [IJCAI 2019]	0.8625	0.8440	0.8660	0.8675	0.8570	0.8555	0.8319	0.8563	0.8603	0.8495	0.8080	0.7717	0.8125	0.8141	0.8133
	EEMIVC [TPAMI 2021]	0.9430	0.9330	0.8995	0.9020	0.8875	0.8810	0.8628	0.8007	0.8028	0.7517	0.8788	0.8578	0.7921	0.7949	0.7329
	IMVC-CBG [CVPR 2022]	0.8885	0.9190	0.8905	0.8855	0.8655	0.8070	0.8400	0.7919	0.7872	0.7848	0.7733	0.8309	0.7740	0.7675	0.7635
	SIMVC-SA [MM 2023]	0.8780	0.8890	0.8130	0.8200	0.8315	0.7689	0.7936	0.6905	0.6828	0.6982	0.7510	0.7736	0.6438	0.6472	0.6687
	FCMVC-IVA [TIP 2024]	0.7515	0.6735	0.6230	0.4960	0.3610	0.6254	0.5238	0.4521	0.3530	0.2800	0.5610	0.4452	0.3601	0.2389	0.1559
	FMVC-VIA [TNNLS 2024]	0.9325	0.9230	0.9115	0.8925	0.8985	0.8591	0.8455	0.8219	0.7899	0.8025	0.8572	0.8392	0.8147	0.7789	0.7094
	FSIMVC-OF [MM 2024]	0.9530	0.9450	0.9330	0.9305	0.9165	0.8999	0.8944	0.8666	0.8573	0.8396	0.8990	0.8875	0.8603	0.8531	0.8287
	LFIMVC [TIP 2024]	0.8805	0.8535	0.8310	0.7885	0.8200	0.7876	0.7518	0.7280	0.6769	0.6827	0.7589	0.7180	0.6820	0.6252	0.6537
	PSIMVC-PG [TNNLS 2024]	0.8335	0.8950	0.8445	0.8435	0.7760	0.7810	0.8056	0.7466	0.7406	0.6779	0.7199	0.7849	0.7100	0.6981	0.6184
	CAL [IJCAI 2025]	0.8615	0.7805	0.6825	0.5965	0.5975	0.8850	0.7883	0.6572	0.5698	0.5622	0.8309	0.6664	0.3983	0.2590	0.3154
	CARD (Ours)	0.9580	0.9615	0.9375	0.9410	0.9280	0.9075	0.9144	0.8730	0.8721	0.8547	0.9101	0.9168	0.8837	0.8752	0.8458

